

mb-pc40s-2025-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

sill

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

Solution

$$\frac{1}{R} \cdot \underline{10} \cdot \underline{18} \cdot \underline{10} \cdot \underline{18} \cdot \underline{10} = 324000$$

Note:

The restriction of the first letter does not need to be shown.

Question 2

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where $R(t)$ is the percentage, in decimal form, of people who click on the link
 t is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

Solution

$$0.45 = -(0.8)^{0.2t} + 0.6$$

$$-0.15 = -(0.8)^{0.2t}$$

$$0.15 = (0.8)^{0.2t}$$

$$\log(0.15) = \log(0.8)^{0.2t}$$

$$\log(0.15) = 0.2t \cdot \log(0.8)$$

$$\frac{\log(0.15)}{0.2 \cdot \log(0.8)} = t$$

$$t = 42.508958\dots$$

$$t = 42.509 \text{ days}$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$\frac{1}{2}$ mark for evaluating quotient of logarithms

2 marks

Question 3

Determine and simplify the middle term in the binomial expansion of $\left(\frac{5}{x} + 4x^3\right)^6$.

Solution

$$\begin{aligned}t_4 &= {}_6C_3 \left(\frac{5}{x}\right)^3 (4x^3)^3 \\&= 20 \left(\frac{125}{x^3}\right) (64x^9) \\&= 160000x^6\end{aligned}$$

2 marks (1 mark for ${}_6C_3$; $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Question 4

Solve $3 \tan^2 x + 5 \tan x - 6 = 0$, over the interval $[0, 2\pi]$.

Solution

$$\tan x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\tan x = \frac{-5 \pm \sqrt{97}}{6} \quad 1 \text{ mark for solving for } \tan x$$

$$\tan x = 0.808142\dots$$

$$x_r = 0.679686\dots$$

$$x = 0.680, 3.821$$

$$\tan x = -2.474809\dots$$

$$x_r = 1.186784\dots$$

$$x = 1.955, 5.096$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

3 marks

Question 5

Given $\theta = -80^\circ$,

a) determine the measure of θ , in radians.

b) state all the coterminal angles of θ .

Solution

a)

$$\theta = -80^\circ \left(\frac{\pi}{180^\circ} \right)$$

$$\theta = -\frac{4\pi}{9}$$

or

$$\theta = -1.396$$

b)

$$\theta = -80^\circ + 360^\circ k, k \in \mathbb{Z}$$

or

$$\theta = -\frac{4\pi}{9} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta = -1.396 + 2k\pi, k \in \mathbb{Z}$$

Question 6

State a possible value of n if the polynomial function $p(x) = (x + 1)^n(x - 4)^2$ has a range of $(-\infty, \infty)$.

Solution

$$n = 1$$

or

1 mark

$$n = 3$$

Note:

Accept any odd whole number for n .

Question 7

Describe the transformations used to obtain the graph of the function $y = -2f(x + 8) - 5$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is reflected over the x -axis, vertically stretched by a factor of 2, translated left 8 units and down 5 units.

1 mark for vertical reflection

1 mark for vertical stretch

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Note:

Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

Question 8

State the domain and range of the function $f(x) = \frac{-2}{x^2}$.

Domain:

Range:

Solution

Domain: $\{x \mid x \neq 0, x \in \mathbb{R}\}$ or $(-\infty, 0) \cup (0, \infty)$ 1 mark for domain

Range: $\{y \mid y < 0, y \in \mathbb{R}\}$ or $(-\infty, 0)$ 1 mark for range

2 marks

Question 9

Given the following row of Pascal's Triangle, state the values in the next row.

1 5 10 10 5 1

Solution

1615201561

Question 10

Given $f(x) = 3 - 2x$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = 3 - 2x$$

To determine the inverse of $f(x)$, switch x and y .

$$x = 3 - 2y$$

$$2y = 3 - x$$

$$y = \frac{3 - x}{2}$$

$$f^{-1}(x) = \frac{3 - x}{2}$$

1 mark for switching x and y

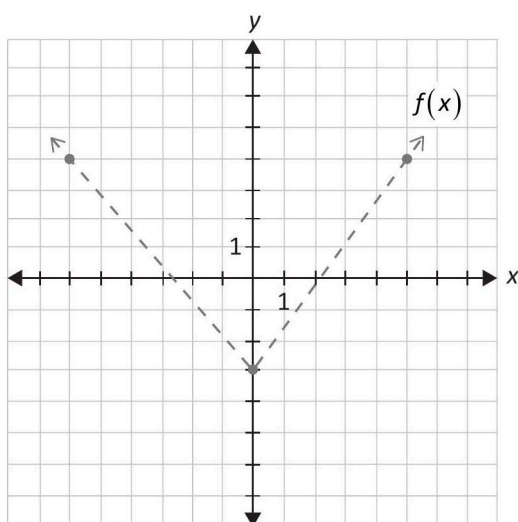
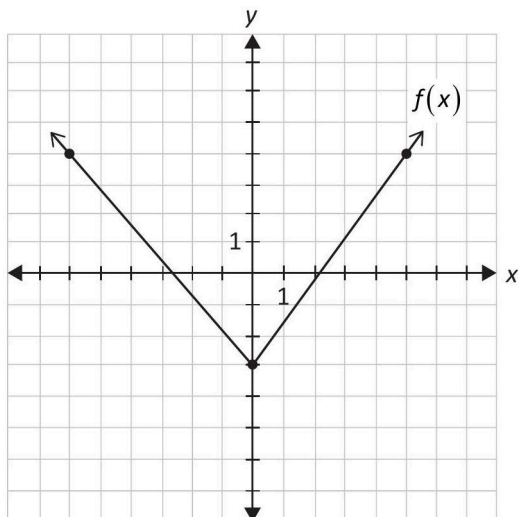
$\frac{1}{2}$ mark for solving for y

$\frac{1}{2}$ mark for writing equation in terms of $f^{-1}(x)$

2 marks

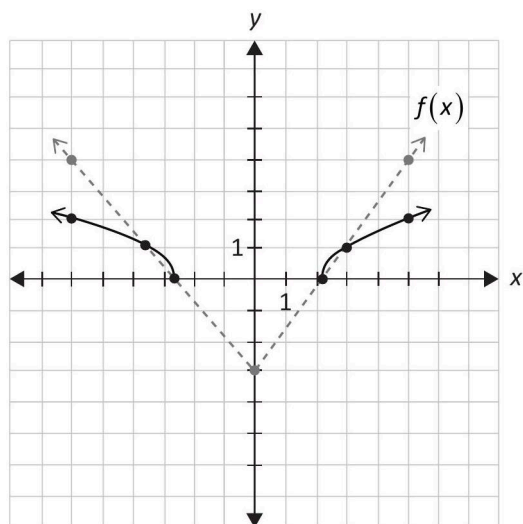
Question 11

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of
 $f(x)$ has already
been drawn for
your reference.
No marks will be
awarded for the
graph of $f(x)$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant
points, $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points,
 $\{y \geq 1\}$

2 marks

Question 12

Prove the identity for all permissible values of x .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$ $\frac{\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}}{\frac{\cos^2 x + \sin^2 x}{\cos x \sin x}}$ $\frac{\cos x \sin x}{\sin^2 x \cos^2 x}$ $\frac{1}{\sin x \cos x}$ $\csc x \sec x$	$\csc x \sec x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

Justify that the graph of $f(x) = e^{x+1} + 5$ does not have an x -intercept.

Solution

$$\begin{aligned}0 &= e^{x+1} + 5 \\ -5 &= e^{x+1}\end{aligned}$$

$e^{x+1} > 0$, therefore there is no solution.

or

The range of the graph is $\{y \mid y > 5, y \in \mathbb{R}\}$.

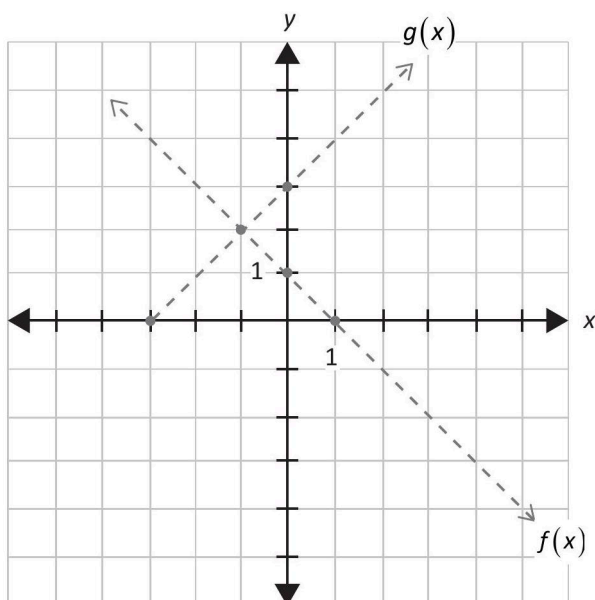
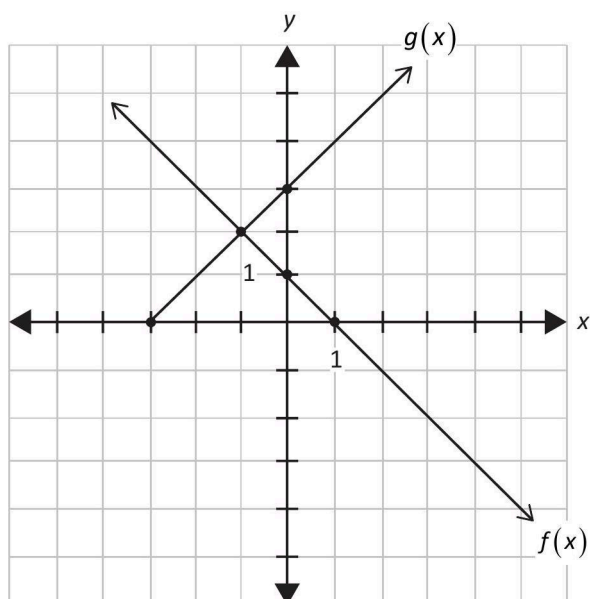
or

The horizontal asymptote is $y = 5$ and it is an increasing exponential function above the asymptote.

1 mark

Question 14

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) \bullet g(x)$.



The graph of $f(x)$

and $g(x)$ have

already been

drawn for your

reference.

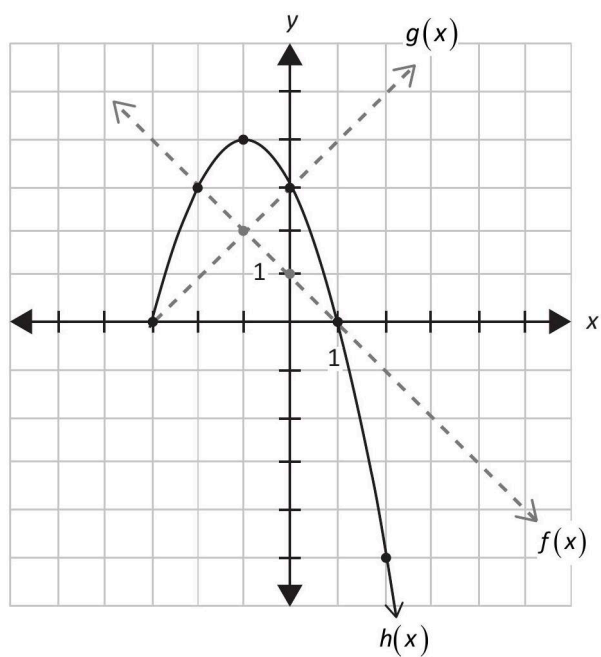
No marks will be

awarded for the

graph of $f(x)$

and $g(x)$.

Solution



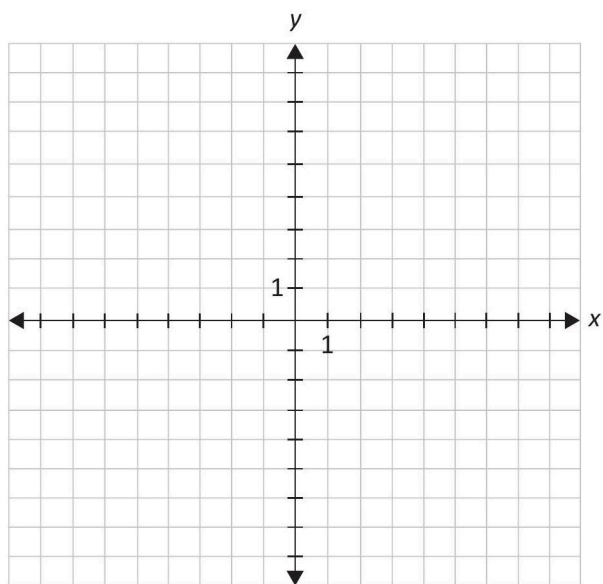
1 mark for operation of multiplication

1 mark for restricted domain

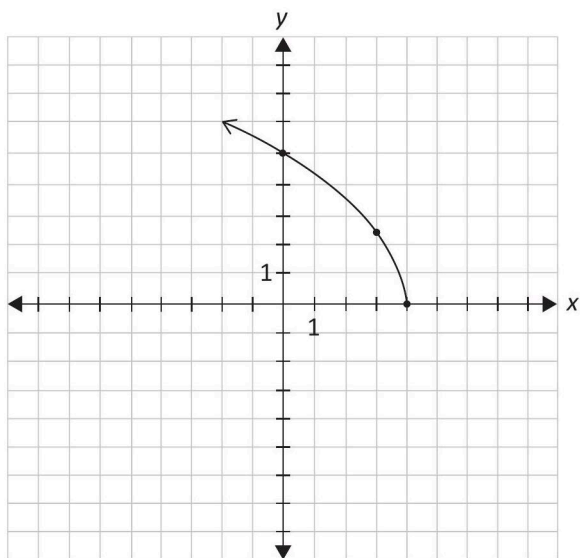
2 marks

Question 15

Sketch the graph of $f(x) = \frac{5}{2}\sqrt{-(x-4)}$.



Solution



1 mark for shape of a radical function

1 mark for vertical stretch

1 mark for horizontal reflection

1 mark for horizontal translation

4 marks

Question 16

Identify the number of x -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

A.

1

B.

2

C.

3

D.

4

Solution

Correct answer: B

2

Question 17

Identify the equation that represents the graph of $y = x^2 - 3$ after a reflection over the x -axis.

A.

$$y = -x^2 - 3$$

B.

$$y = x^2 - 3$$

C.

$$y = x^2 + 3$$

D.

$$y = -x^2 + 3$$

Solution

Correct answer: D

$$y = -x^2 + 3$$

Question 18

Given $2^{\log_2 3} = x$, identify the value of x .

A.

1

B.

3

C.

6

D.

8

Solution

Correct answer: B

3

Question 19

Identify the y -intercept on the graph of $y = \frac{x-3}{x^2-5x+6}$.

A.

−3

B.

−2

C.

− $\frac{1}{2}$

D.

3

Solution

Correct answer: C

− $\frac{1}{2}$

Question 20

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

A.

$10!2!$

B.

$11!$

C.

$11!2!$

D.

$12!$

Solution

Correct answer: C

$11!2!$

Question 21

Identify which expression represents the remainder when the polynomial $P(x)$ is divided by $(x - 4)$.

A.

$$P(-4)$$

B.

$$P(4)$$

C.

$$P(x - 4)$$

D.

$$P(x + 4)$$

Solution

Correct answer: B

$$P(4)$$

Question 22

Identify the period of the graph of the sinusoidal function.

A.

10π

B.

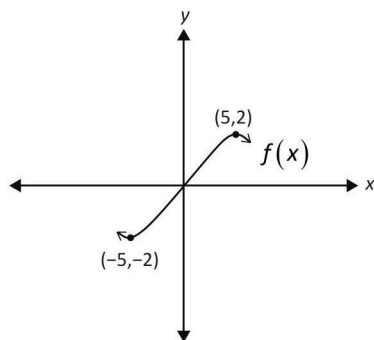
10

C.

20π

D.

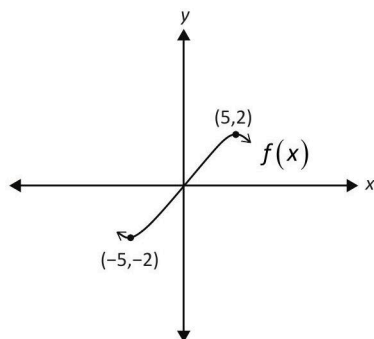
20



Solution

Correct answer: D

20



Question 23

1 mark

Identify an equivalent expression for $\log_3 9 + \log_3 5$.

A.

$$\log_3 14$$

B.

$$2 \log_3 5$$

C.

$$\log_3 45$$

D.

$$\log_3 \left(\frac{9}{5} \right)$$

Solution

Correct answer: C

$$\log_3 45$$

Question 24

Identify the coterminal angle(s) of $\theta = -\frac{11\pi}{10}$ over the interval $[-3\pi, \pi]$.

A.

$$\theta = \frac{9\pi}{10}$$

B.

$$\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$$

C.

$$\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$$

D.

$$\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$$

Solution

Correct answer: A

$$\theta = \frac{9\pi}{10}$$

Question 25

Express $p(x) = x^3 - 13x - 12$ in completely factored form.

$$p(x) =$$

Solution

$$p(-1) = (-1)^3 - 13(-1) - 12$$

$$p(-1) = 0$$

$\therefore (x + 1)$ is a factor

1 mark for identifying one possible zero of $P(x)$

-1	1	0	-13	-12
		-1	1	12
	1	-1	-12	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x + 1)(x^2 - x - 12)$$

$$p(x) = (x + 1)(x - 4)(x + 3)$$

1 mark for consistent product of factors

3 marks

Question 26

Verify, by substitution, that the equation $\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{2 \csc \theta}{\sec^2 \theta}$ is true for $\theta = \frac{\pi}{6}$.

Left-Hand Side	Right-Hand Side

Solution

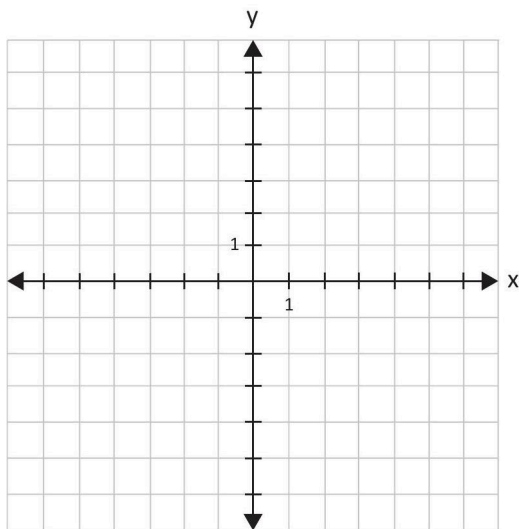
Left-Hand Side	Right-Hand Side
$\frac{\cos^2\left(\frac{\pi}{6}\right)}{\sin^2\left(\frac{\pi}{6}\right)}$ $\frac{\left(\frac{\sqrt{3}}{2}\right)^2}{\left(\frac{1}{2}\right)^2}$ $\frac{3}{4} \div \frac{1}{4}$	$\frac{2 \csc\left(\frac{\pi}{6}\right)}{\sec^2\left(\frac{\pi}{6}\right)}$ $\frac{2(2)}{\left(\frac{2}{\sqrt{3}}\right)^2}$ 2 marks for exact values ($\frac{1}{2}$ mark for each) $\frac{2(2)}{\left(\frac{4}{3}\right)}$ $\frac{12}{4}$
3	3 1 mark for simplification ($\frac{1}{2}$ mark for LHS; $\frac{1}{2}$ mark for RHS) 3 marks

Question 27

Given $f(x) = \log_3 x + 2$,

a) determine the x -intercept of $f(x) = \log_3 x + 2$.

b) sketch the graph of $f(x) = \log_3 x + 2$.



Solution

a)

$$0 = \log_3 x + 2$$

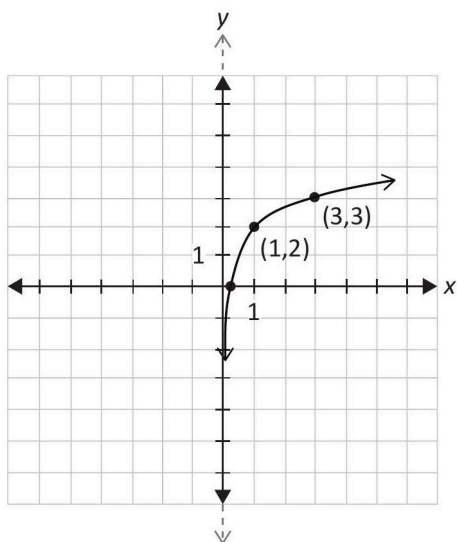
$$-2 = \log_3 x$$

$$3^{-2} = x$$

$$\frac{1}{3^2} = x$$

$$\frac{1}{9} = x$$

b)



1 mark for shape of a logarithmic function

1 mark for vertical translation

2 marks

Question 28

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

Solution

$$\left(-\frac{2}{\sqrt{2}}\right)^2 \left(\frac{\sqrt{3}}{1}\right)^2$$

1 mark for $\sec\left(\frac{5\pi}{4}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

$$\left(\frac{4}{2}\right)\left(\frac{3}{1}\right)$$

6

1 mark for $\tan\left(-\frac{2\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

2 marks

Question 29

Given the functions $f(x) = x - 3$ and $g(x) = \sqrt{x} + 2$, state the domain of $g(f(x))$.

Solution

$$\{x \mid x \geq 3, x \in \mathbb{R}\}$$

1 mark

or

$$[3, \infty)$$

Question 30

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

Solution

$$\begin{aligned}\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ &= \sin(70^\circ - 25^\circ) \\ &= \sin 45^\circ \\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

$\frac{1}{2}$ mark for substitution into correct identity

$\frac{1}{2}$ mark for exact value

1 mark

Question 31

Solve, algebraically.

$$\log_{15}(x^2 - 1) = 1$$

Solution

Method 1

$$\begin{aligned}15^1 &= x^2 - 1 \\0 &= x^2 - 16 \\0 &= (x + 4)(x - 4) \\x &= -4, x = 4\end{aligned}$$

1 mark for exponential form

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

2 marks

Method 2

$$\begin{aligned}\log_{15}(x^2 - 1) &= \log_{15} 15 \\x^2 - 1 &= 15 \\x^2 &= 16 \\x &= \pm 4\end{aligned}$$

$\frac{1}{2}$ mark for logarithmic form

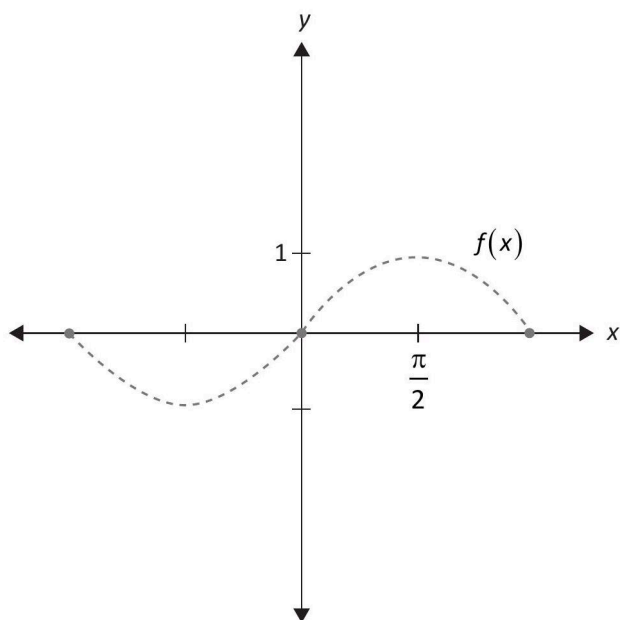
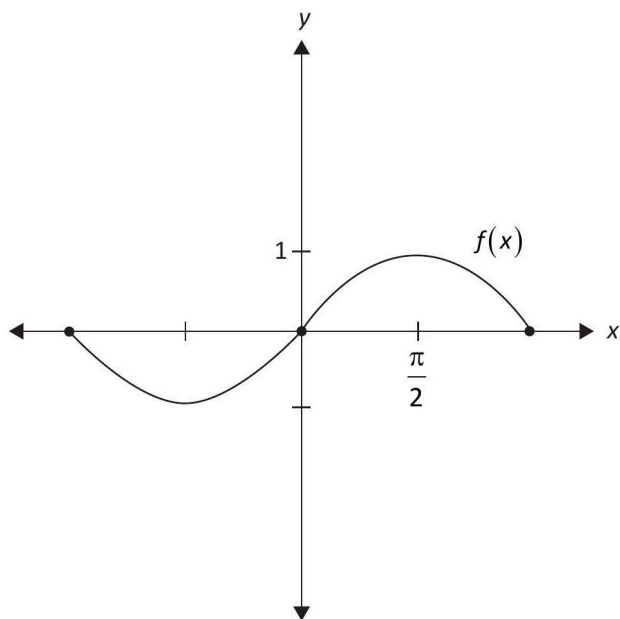
$\frac{1}{2}$ mark for equating arguments

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

2 marks

Question 32

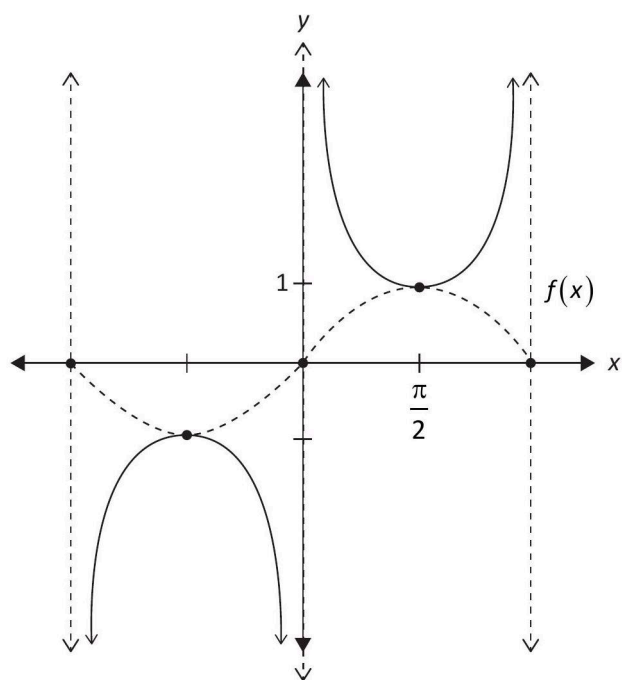
Given the graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.



The graph of $f(x)$
has already been
drawn for your
reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution



1 mark for asymptotic behaviour approaching $x = -\pi$, $x = 0$, and $x = \pi$

1 mark for invariant points ($\frac{1}{2}$ mark for $y = -1$; $\frac{1}{2}$ mark for $y = 1$)

2 marks

Question 33

A light is attached to the end of a blade of a wind turbine.

The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

where $h(t)$ is the height of the light with respect to the ground, measured in feet and,
 t is the time, measured in seconds.



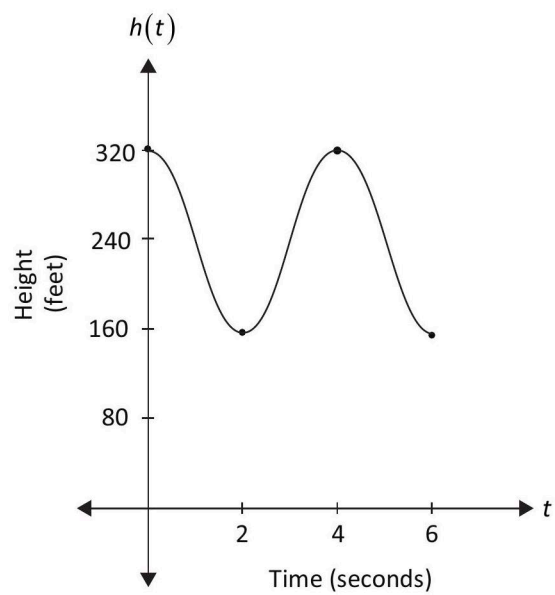
- a) Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.



- b) If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of $h(t)$.

Solution

a)



b) The period would decrease.

Question 34

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628800$$

Describe her error.

Solution

She did not consider the repeating letters.

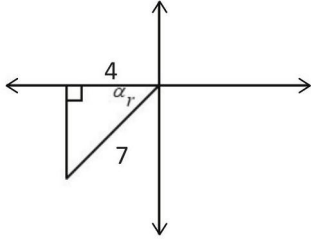
Question 35

Given that $\cos \alpha = -\frac{4}{7}$ where α is in quadrant III and $\sin \beta = \frac{5}{13}$ where β is in quadrant II, determine the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sec(\alpha + \beta)$

Solution



$$\leftarrow \sqrt{13} \beta_r$$

$$x^2 + y^2 = r^2$$

$$16 + y^2 = 49$$

$$y^2 = 33$$

$$y = \pm\sqrt{33}$$

$$x^2 + y^2 = r^2$$

$$x^2 + 25 = 169$$

$$x^2 = 144$$

$$x = \pm 12$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(-\frac{4}{7}\right)\left(-\frac{12}{13}\right) - \left(-\frac{\sqrt{33}}{7}\right)\left(\frac{5}{13}\right)$$

$$= \frac{48 + 5\sqrt{33}}{91}$$

$\frac{1}{2}$ mark for consistent value of $\cos \beta$

$\frac{1}{2}$ mark for consistent value of $\sin \alpha$

3 marks

1 mark for substitution into correct identity

b) $\sec(\alpha + \beta) = \frac{91}{48 + 5\sqrt{33}}$

1 mark for consistent value of $\sec(\alpha + \beta)$

1 mark

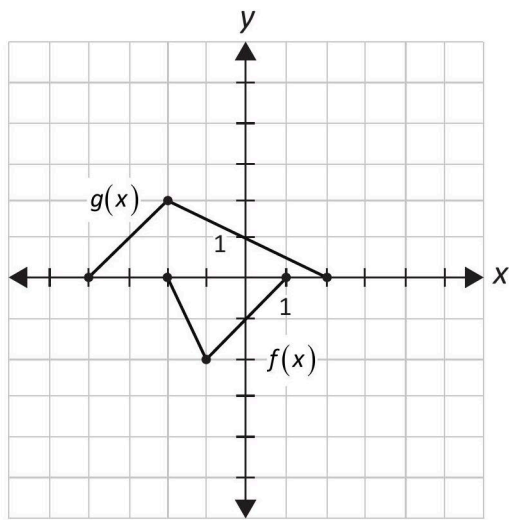
Note:

Accept any of the following values for x : $x = \pm 12$, $x = -12$, or $x = 12$.

Accept any of the following values for y : $y = \pm\sqrt{33}$, $y = -\sqrt{33}$, or $y = \sqrt{33}$.

Question 36

Determine the equation for $g(x)$, in terms of $f(x)$.



$g(x) =$

Solution

$$g(x) = -f\left(\frac{x}{2}\right)$$

1 mark for vertical reflection

1 mark for horizontal stretch

2 marks

Question 37

State possible values for m and n that satisfy the equation, $\log_m n = 5$.

Solution

Some possible solutions are:

$$m = 2 \quad n = 32$$

or

$$m = 3 \quad n = 243$$

1 mark

or

$$m = 4 \quad n = 1024$$

or

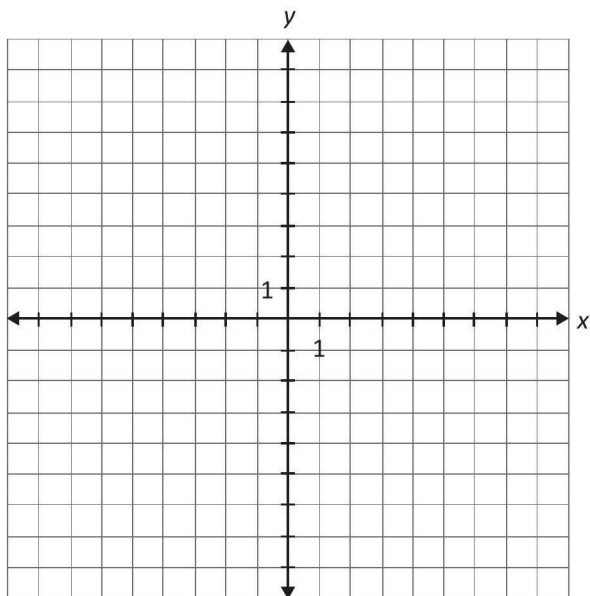
$$m = 5 \quad n = 3125$$

Notes:

Other values for m and n are possible.

Question 38

Sketch the graph of $f(x) = \frac{x-4}{x^2-7x+12}$.

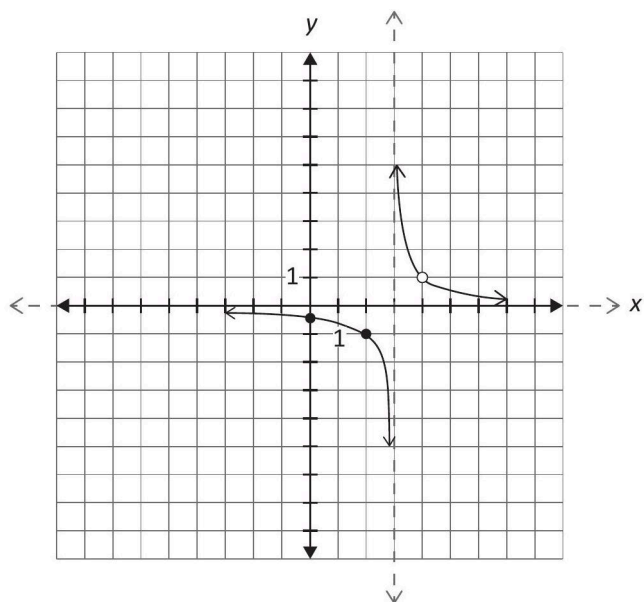


Solution

$$\begin{aligned} f(x) &= \frac{(x-4)}{(x-4)(x-3)} \\ &= \frac{1}{x-3}, x \neq 4 \end{aligned}$$

$\therefore$ there is a point of discontinuity

(hole) at $(4, 1)$



1 mark for asymptotic behaviour approaching $x = 3$

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for point of discontinuity (hole) at $x = 4$

1 mark for shape of rational function ($\frac{1}{2}$ mark for graph left of $x = 3$, $\frac{1}{2}$ mark for graph right of $x = 3$)

4 marks

Notes:

Deduct $\frac{1}{2}$ mark for procedural error (incorrect y -value for point of discontinuity (hole)).

Question 39

Justify that the value of ${}_3P_4$ does not exist.

Solution

Method 1

When using

$${}_nP_r, n \geq r$$

.

Method 2

$${}_3P_4 = \frac{3!}{(3-4)!}$$

$${}_3P_4 = \frac{3!}{(-1)!}$$

$(-1)!$ does not exist, therefore

$${}_3P$$

does not exist.

1 mark

Question 40

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at $(\frac{\pi}{2}, 1)$

Solution

$$\{y \mid -5 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[-5, 1]$$

1 mark

Question 41

Solve, algebraically.

$${}_nC_2 = 15$$

Solution

$$\frac{n!}{2!(n-2)!} = 15$$

$$\frac{n(n-1)(n-2)!}{2!(n-2)!} = 15$$

$$n(n-1) = 30$$

$$n^2 - n - 30 = 0$$

$$(n-6)(n+5) = 0$$

$$n = 6 \quad n \geq -5$$

$\frac{1}{2}$ mark for substitution

1 mark for factorial expansion

$\frac{1}{2}$ mark for simplification of factorials

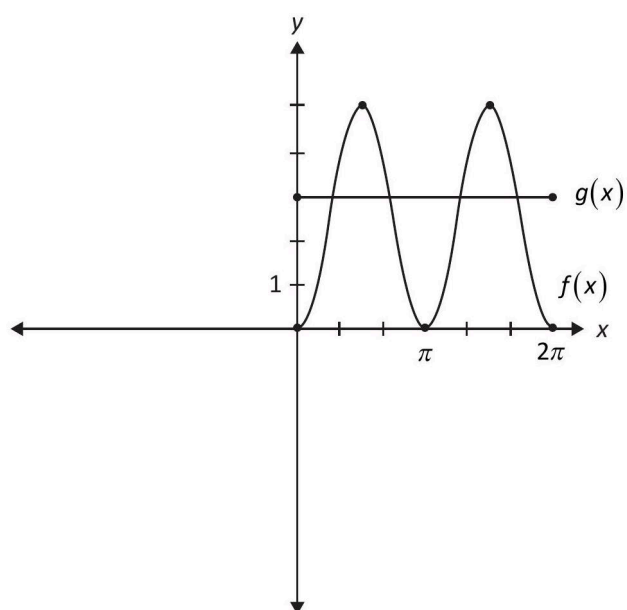
$\frac{1}{2}$ mark for the permissible value of n

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Question 42

Describe how to use the graphs of $f(x) = 5 \sin^2 x$ and $g(x) = 3$ to solve the equation, $5 \sin^2 x = 3$, over the interval $[0, 2\pi]$.



Solution

The solution will be the x -values where the two graphs intersect.

1 mark

Question 43

Solve, algebraically.

$$2 \log_3 5 - \frac{1}{3} \log_3 125 = \log_3 a$$

Solution

Method 1

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} = \log_3 a$$

$$\log_3 25 - \log_3 5 = \log_3 a$$

$$\log_3 \left(\frac{25}{5} \right) = \log_3 a$$

$$\log_3(5) = \log_3 a$$

$$a = 5$$

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

1 mark for equating arguments

3 marks

Method 2

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} - \log_3 a = 0$$

$$\log_3 \left(\frac{25}{5a} \right) = 0$$

$$3^0 = \frac{25}{5a}$$

$$5a = 25$$

$$a = 5$$

1 mark for power law ($\frac{1}{2}$ mark for each)

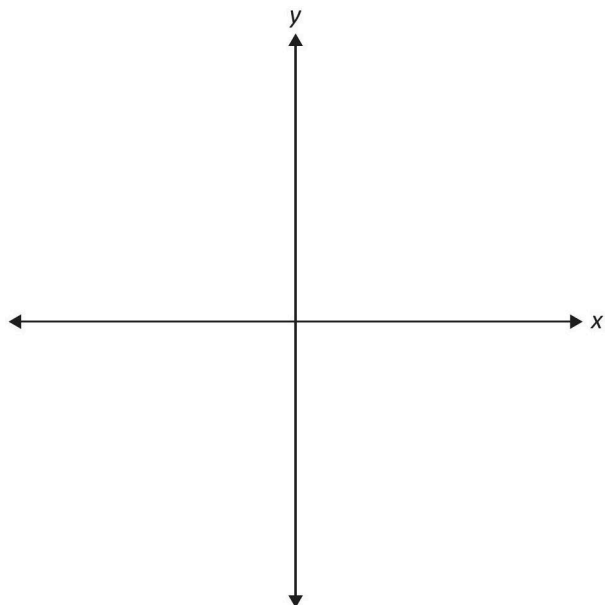
1 mark for quotient law

1 mark for exponential form

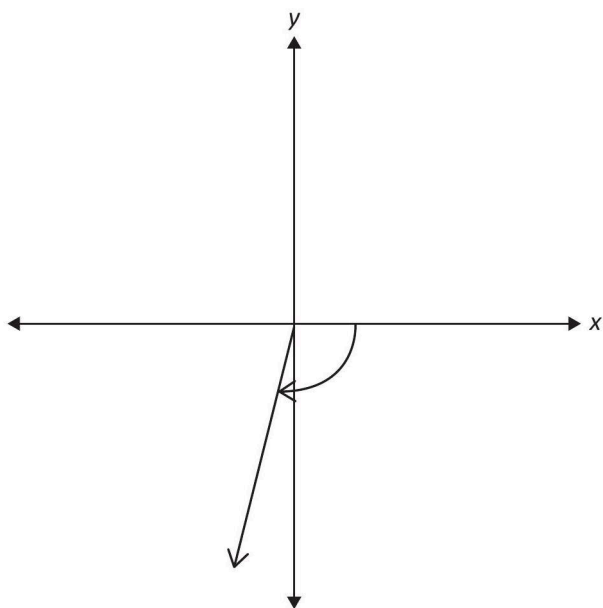
3 marks

Question 44

Sketch the angle, $\frac{-7\pi}{12}$, in standard position.



Solution



$\frac{1}{2}$ mark for an appropriate angle

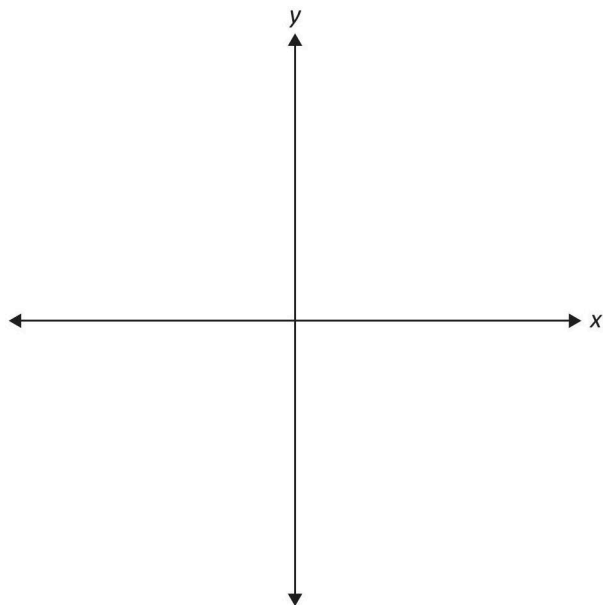
in quadrant III

$\frac{1}{2}$ mark for correct direction

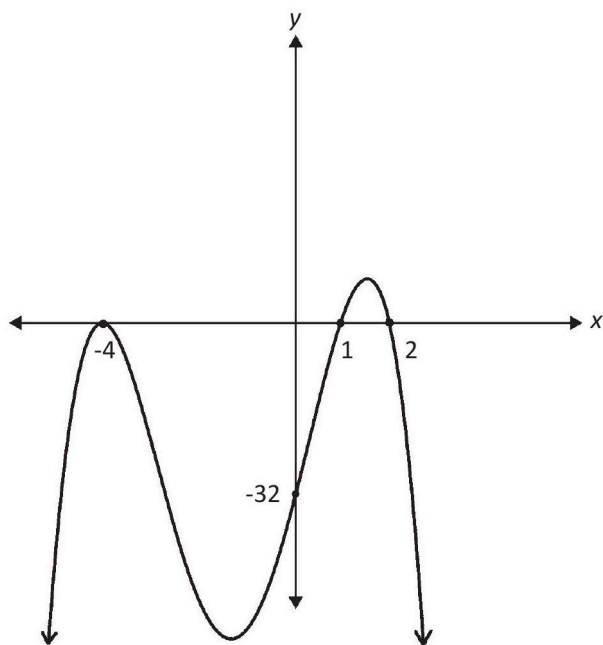
1 mark

Question 45

Sketch the graph of $p(x) = -(x-1)(x-2)(x+4)^2$.



Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -4$

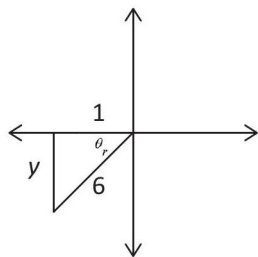
$\frac{1}{2}$ mark for end behaviour

3 marks

Question 46

The point, $P(\theta) = (-\frac{1}{6}, y)$, lies on the unit circle and is located in Quadrant III. Determine the exact value of $\csc \theta$.

Solution



$$(-1)^2 + y^2 = (6)^2 \quad 1/2 \text{ mark for substitution}$$

$$1 + y^2 = 36$$

$$y^2 = 35 \quad 1/2 \text{ mark for solving for } y$$

$$y = \pm\sqrt{35} \quad 1 \text{ mark for consistent value of } \csc \theta$$

$$\csc \theta = -\frac{6}{\sqrt{35}} \quad (1/2 \text{ mark for quadrant; } 1/2 \text{ mark for value})$$

or

$$\csc \theta = -\frac{6\sqrt{35}}{35}$$

Notes:

Accept any of the following values for y : $y = \pm\sqrt{35}$, $y = \sqrt{35}$, or $y = -\sqrt{35}$.